Appendix IV

ZJU-UIUC INSTITUTE
Zhejiang University-University of Illinois Urbana-Champaign Institute
浙江大学伊利诺伊大学厄巴纳香槟校区联合学院

Literature Review
&
Project Proposal
for
ST 4001
Senior Thesis

Thesis Title	Mixed-Precision Quantization for SNN Inference
Student Name	Ding Jiang
Student ZJU ID	3220110672
Major	Electrical Engineering
Supervisor name & Signature*	Aili Wang

* Supervisor shall sign the proposal if s/he approves the student to submit the proposal to institute.

Table of Contents

Literature Review

1 Introduction 2

1.1 Background info. 2

2 Current Research Fields 3

2.1 Research direction and current progress 3

2.2 Existing problems and/or obstacles 4

3 Prospectives in the Research Field. 5

Project Proposal

1 Research Background 7

1.1 Introduction to the research fields 7

1.2 Aim and significance of the proposal. 8

2 Proposed Research Approaches 9

2.1 Major research approaches 9

2.2 Techniques and methodology 9

2.3 Feasibility analysis 10

3 Project Schedule and Expected Outcome 10

3.1 Research schedule 10

3.2 Expected outcome 12

4 Research Progress 13

4.1 Preliminary results 13

References

Literature Review

1. Introduction

1.1 Background info

With the explosive growth of parameters in artificial intelligence models, traditional deep learning faces severe memory walls and energy consumption bottlenecks on hardware based on the von Neumann architecture. Against this backdrop, spiking neural networks (SNNs), inspired by the biological brain, have attracted significant attention as third-generation neural networks due to their unique computational paradigm. Unlike traditional artificial neural networks (ANNs), which rely on continuous numerical data to transmit information, SNNs use discrete spike trains for communication, exhibiting significant event-driven characteristics and spatio-temporal sparsity. This means that neurons only fire spikes when the membrane potential accumulates above a threshold; during periods without spikes, the network remains silent, consuming almost no dynamic power. Furthermore, SNNs transform the expensive floating-point multiply-accumulate operations (MACs) in ANNs into more efficient accumulation operations (ACs), theoretically achieving several orders of magnitude energy efficiency improvements on neuromorphic hardware [1][2].

However, despite the theoretically high energy efficiency potential of SNNs, they still face significant challenges in practical edge deployments. In pursuit of high-precision task performance, modern SNN models tend to be more complex, resulting in full-precision floating-point weights and membrane potentials occupying a large amount of on-chip storage space, causing significant memory access overhead and bandwidth pressure. To bridge the gap between theoretical energy efficiency and actual hardware resources, **model quantization** technology has become a key solution. By mapping high-precision floating-point parameters to low-bit fixed-point numbers or integers, quantization technology can not only significantly compress the model size, but also further reduce inference latency and energy consumption by simplify-

ing the underlying logic circuits (e.g., completely removing multipliers) [3]. Therefore, how to maintain the complex spatiotemporal dynamics of SNNs while ensuring high-precision inference capabilities under the constraint of extremely low bit width has become a core problem that urgently needs to be solved in the current field of neuromorphic computing.

2. Current Research Fields

2.1 Research direction and current progress

Current research on efficient inference in SNNs mainly revolves around the main themes of extreme uniform quantization, fast inference transformation and exploration of new architectures, aiming to break through performance bottlenecks while compressing the model.

In terms of uniform quantization algorithms, researchers are committed to maximizing the preservation of network information under extremely low bit settings. The MINT method proposed by Yin et al. [3] represents the forefront of this direction. This work innovatively quantizes weights and membrane potentials into 2-bit integers simultaneously, and completely removes the expensive multipliers in the inference stage through the design of shared scaling factors, realizing a pure shift-add network. To further alleviate the accuracy loss caused by quantization, Wei et al. [4] introduced a weight-pulse dual adjustment mechanism based on information entropy in Q-SNNs. By maximizing the information capacity of binary weights and low-bit pulses, they proved that even under extreme compression, SNNs can still maintain high feature representation capabilities. Subsequently, Wei et al. [5] expanded the research dimension from single quantization to joint compression in QP-SNNs. By combining uniform quantization and structured pruning, and proposing a new pruning standard based on spatiotemporal impulse activity, they achieved a dual reduction in model size and computational cost.

Meanwhile, in order to overcome the inference latency problem caused by the long temporal encoding required by SNNs, optimizing the ANN-to-SNN conversion algorithm has become

another important branch. Hu et al. [6] proved the equivalence of ANN spatial quantization and SNN temporal quantization through mathematical derivation in Fast-SNN, and used signed neurons to correct the sequence error, remarkably reducing the inference latency to only 3 time steps. Li et al. [7] proposed a suppression strategy for the ”random noise” problem in the conversion process in their QFFS framework, enabling SNNs to achieve accuracy comparable to direct training in a very short time on complex datasets such as ImageNet. In addition, the research frontier has begun to shift towards more complex Transformer architectures. Yao et al. [8] proposed Spike-driven Transformer, which successfully introduced the spiking mechanism into the self-attention module, removed Softmax and floating-point multiplication, and verified that the quantization advantage of SNN can be extended to large-scale pre-trained models.

2.2 Existing problems and/or obstacles

Although the above studies have made significant progress in single-dimensional optimization, existing SNN quantization frameworks generally adopt a layer-wise uniform strategy. This rigid paradigm gradually reveals its limitations when facing deep networks and complex hardware constraints.

First, existing methods such as MINT [3] and Q-SNNs [4] ignore the differences between layers within the network. In the ANN field, Dong et al. [9]’s HAWQ study has demonstrated through Hessian spectrum analysis that different layers of the network have vastly different sensitivities to quantization noise. However, the current SNN field lacks a similar theoretical framework to distinguish between critical layers and redundant layers, resulting in severe information loss in critical layers and wasted resources in redundant layers when using the same low bit depth for all layers. Second, from the perspective of hardware implementation, uniform quantization is difficult to adapt to the physical constraints of neuromorphic chips. Taking IBM TrueNorth [1] as an example, this chip natively supports extremely low precision weights. If high precision is forced to be simulated, the number of neuron cores will be multiplied. Conversely, although Intel Loihi [2] supports mixed precision, existing algorithms have failed to effectively utilize this feature to balance precision and resource consumption.

Furthermore, the training dynamics of SNNs are also constrained by single precision. Moshruba et al. [10] pointed out in their latest research that reducing the quantization level will significantly change the distribution characteristics of surrogate gradients, leading to gradient confusion. This means that using extremely low bit widths uniformly across all layers will cause the gradient information of key layers to be distorted, thereby hindering the convergence of deep networks. Finally, although Na et al. [11]'s AutoSNN has enabled automated architecture search for SNNs, the current search space has not yet covered the dimension of quantization bit width, and there is a lack of an automated mechanism to find Pareto optimal solutions in the huge space of mixed precision combinations.

3. Prospectives in the Research Field

Based on the analysis and insights into the limitations of existing literature, future SNN inference research will inevitably shift from single uniform quantization to a more flexible and efficient mixed-precision quantization. This paradigm shift will no longer treat the network as a homogeneous whole, but will intelligently allocate resources based on the contribution of each layer to the final task accuracy.

Future research first needs to establish a sensitivity analysis theory applicable to the spatiotemporal dynamics of SNNs. Drawing on the idea of HAWQ [9], sensitive layers in the network can be accurately located by analyzing the Hessian matrix or gradient information. A higher bit width is retained in these key layers to maintain the accuracy of the gradient flow (solving the gradient distortion problem proposed in [10]), while aggressive compression is performed on non-sensitive layers, thereby breaking through the performance ceiling of existing uniform quantization methods such as MINT [3]. Secondly, the mixed-precision strategy will be deeply coupled with hardware characteristics. By leveraging the native support for multi-precision in chips such as Loihi [2] and designing specific cost functions for resource-constrained platforms such as TrueNorth [1], the research will focus on finding the optimal balance between model accuracy and hardware costs (such as neuron occupancy and energy consumption). Finally,

in order to cope with the huge search space, it will be inevitable to extend the framework of AutoSNN [11] to support automated bit-width search. By introducing differentiable search or evolutionary algorithms, future work is expected to automatically discover hybrid bit architectures that balance low latency (refer to the Fast-SNN [6] standard) and high precision, and promote the large-scale application of spiking neural networks on real-world edge intelligent devices.

Project Proposal

1. Research Background

1.1 Introduction to the research fields

As artificial intelligence applications migrate to edge and mobile devices, the high energy consumption of traditional deep learning models on the von Neumann architecture has become a key bottleneck restricting their development. As the core paradigm of neuromorphic computing, spiking neural networks (SNNs) can theoretically provide an energy efficiency ratio several orders of magnitude higher than artificial neural networks (ANNs) by mimicking the event-driven mechanism and spatiotemporal sparsity of the biological brain. However, modern SNNs often adopt deep architectures in pursuit of high precision, resulting in full-precision floating-point weights and membrane potentials occupying a huge amount of on-chip storage space, which is in sharp contradiction with the extremely limited memory resources of edge devices. Therefore, model quantization technology has emerged to compress models by mapping high-precision parameters to low-bit data. Although existing works such as MINT [3] and Q-SNNs [3] have successfully compressed SNNs to low bits, how to maintain the complex spatiotemporal dynamics of the network while performing extreme compression remains a challenge.

1.1.1 Specific subsection

Despite significant progress in existing quantization SNN methods, their core limitation lies in the widespread adoption of rigid uniform quantization strategies. Mainstream methods such as MINT [3] and QP-SNNs [5] force each layer in the network to use the same bit width (e.g., all 2-bit), and this ”one-size-fits-all” approach seriously ignores the heterogeneity within deep networks. According to the theoretical analysis of HAWQ [9] in the field of ANN, different

layers of the network have significantly different sensitivities to quantization noise. Usually, the first and last layers contain rich feature information or classification logic and are extremely sensitive to accuracy, while the middle layers have high redundancy. Uniform quantization thus falls into a dilemma: if a high bit width is used uniformly to protect sensitive layers, it will lead to a waste of a lot of storage resources in redundant layers, making it difficult to meet the stringent constraints of hardware such as TrueNorth [1]; if a low bit width is used uniformly for compression, it will lead to serious loss of sensitive layer information and irreversible decrease in accuracy. In addition, Moshruba et al. [10] showed that extremely low bit quantization can cause gradient confusion, and a unified low bit strategy may cause the gradient flow of key layers to be distorted during backpropagation, thereby hindering the convergence of deep networks.

1.2 Aim and significance of the proposal

To address the aforementioned limitations, this study aims to propose a mixed-precision quantization framework for SNN inference. Its core objective is to break through the performance ceiling of uniform quantization and achieve Pareto optimality between model accuracy and hardware efficiency. Specifically, this study will establish a hierarchical sensitivity evaluation system suitable for SNN spiking dynamics and design an automated bit-width search strategy to intelligently allocate the precision of each layer under given hardware resource constraints. The significance of this study lies not only in the innovation at the algorithm level but also in its significant practical hardware value. It will enable SNNs to fully utilize the native mixed-precision feature of Intel Loihi [2] or achieve efficient deployment of large models on resource-constrained chips such as IBM TrueNorth [1] through fine-grained resource allocation, thereby truly promoting the application of spiking neural networks in low-power scenarios such as autonomous driving and wearable devices.

2. Proposed Research Approaches

2.1 Major research approaches

This research will follow a systematic research path of metrics-search-training to construct a closed-loop hybrid precision optimization system. First, a sensitivity analysis module will be developed to quantitatively evaluate the contribution of each layer in the network to the final inference accuracy, thus distinguishing between sensitive and redundant layers. Second, based on the sensitivity index, a constrained bit-width allocation optimizer will be introduced. This optimizer can automatically search for the optimal bit-width combination for each layer within the constraints of the total model size or energy budget. Finally, to address the training instability introduced by hybrid precision, a hybrid precision fine-tuning framework will be designed to adaptively adjust the surrogate gradient function of layers with different bit widths, ensuring robust convergence of the network during quantization-aware training (QAT).

2.2 Techniques and methodology

In terms of specific technical implementation, this study will first draw on the theoretical foundation of HAWQ [9] to develop an SNN sensitivity measurement method based on the Hessian matrix trace. Considering the non-differentiable nature of SNN, we will use the smoothed surrogate gradient to approximate the calculation of the Hessian trace. The layer with the larger trace value will be judged to be more sensitive to quantization noise and thus be assigned a higher bit width (such as 8-bit), and vice versa. In the bit width allocation stage, this study will model this problem as an integer linear programming (ILP) or use an evolutionary algorithm to solve it. The goal is to maximize the model accuracy under the premise of satisfying the hardware resource constraints of TrueNorth [1] or Loihi [2]. In addition, in response to the gradient distortion problem caused by quantization level pointed out by Moshruba et al. [10], we will design a bit-width-aware surrogate gradient. This technology dynamically adjusts the smooth-

ing factor of the proxy gradient based on the quantization accuracy of each layer. For low-bit layers, the smoothing range is appropriately increased to compensate for gradient truncation errors, thereby ensuring the accuracy of the gradient flow during backpropagation.

2.3 Feasibility analysis

2.3.1 Details on feasibility

This study is highly feasible in terms of theory, tools, and hardware. From a theoretical perspective, mixed-precision quantization has a solid theoretical foundation in the field of ANN, and works such as Fast-SNN [6] have proven the effectiveness of transferring ANN quantization theory to SNN, which provides reliable theoretical support for this project. From a tool perspective, existing open-source SNN frameworks such as SpikingJelly provide complete automatic differentiation interfaces and PyTorch-based backend support, which can greatly reduce the development threshold of sensitivity analysis and mixed-precision training. From a hardware perspective, the Intel Loihi [2] chip explicitly supports flexible weight configuration from 1 to 9 bits, which means that the algorithm proposed in this study is not just limited to the simulation stage, but has the potential to be directly deployed on advanced neuromorphic hardware and generate practical application value.

3. Project Schedule and Expected Outcome

3.1 Research schedule

3.1.1 Phase 0: Setup & Baselines (Nov 2025 – Dec 2025)

This phase will focus on environment setup and benchmark reproduction. The core task is to configure the development environment based on the SpikingJelly framework and reproduce

the performance of state-of-the-art (SOTA) models such as MINT, VGG, and ResNet on static datasets (CIFAR-10/100) and neuromorphic datasets (N-MNIST, DVS-CIFAR10). The ultimate goal is to establish a stable and reproducible training script and ensure that the accuracy of the obtained benchmark model is consistent with the reported data in the corresponding top conference papers, thus establishing a solid baseline for subsequent comparative experiments.

3.1.2 Phase 1: Probing and Sensitivity Analysis (Dec 2025 – Jan 2026)

This phase will focus on multi-granularity quantization sensitivity theoretical analysis of SNNs. The plan is to develop metrics based on Hessian spectrum or Fisher information to explore the network's sensitivity to quantization noise at the parameter-level, block-level, and time-step level, and finally produce detailed sensitivity heatmaps to quantitatively reveal the intrinsic correlation between layer bit width distribution and the model's final inference accuracy.

3.1.3 Phase 2: Mixed-Precision Design (Jan 2026 – Feb 2026)

This phase focuses on the design of mixed-precision algorithms and automated bit-width search. The core work is to model the bit-width allocation problem as a mathematical optimization problem and use integer linear programming (ILP) or greedy search algorithms to search for the optimal bit-width configuration of each layer under strict hardware constraints (such as on-chip memory size limit and total power consumption budget). The ultimate goal is to complete the coding implementation of the search algorithm and obtain the optimal bit-width mapping table of the target model under different resource constraints.

3.1.4 Phase 3: Quantization-Aware Training (QAT) (Feb 2026 – Mar 2026)

This phase will perform quantization-aware training (QAT) based on the configuration obtained in Phase 2 to restore model performance. The focus is on solving the gradient mismatch problem caused by the difference in quantization levels under mixed precision settings. The trained

model will be comprehensively compared with the full precision (FP32) baseline and the post-trained quantization (PTQ) method. The ultimate goal is to obtain a high-performance mixed precision model and ensure that the difference in accuracy between it and the FP32 baseline is controlled within a very small range of 1.0-1.5%.

3.1.5 Phase 4: Hardware Mapping and Evaluation (Mar 2026 – Apr 2026)

This phase will involve hardware-level efficiency estimation and evaluation. The plan is to map the trained mixed-precision model to a hypothetical SNN accelerator model (referencing the hardware constraints of Intel Loihi or IBM TrueNorth). Key hardware metrics will be estimated through simulation, including the number of accumulation operations (AC), energy consumption, and on-chip memory usage. The final output will be a quantitative analysis report, clearly demonstrating the advantages of the mixed-precision solution in the ”energy consumption/resource usage vs. precision” trade-off curve through data charts.

3.1.6 Phase 5: Writing & Repo Packaging (May 2026)

This phase marks the project’s final stage. The main tasks include documenting and packaging the entire research process, building a reproducible codebase containing clear instructions such as *reproduce_baselines* and *reproduce_quant*, completing the writing, revision, and finalization of the graduation thesis, and preparing the final defense materials to ensure that all research results meet the standards for academic publication and archiving.

3.2 Expected outcome

Upon completion, this project is expected to deliver a complete SNN mixed-precision quantization solution and related academic results. First, a modular codebase based on SpikingJelly will be produced, providing comprehensive capabilities from sensitivity analysis and ILP-based automated bit-width search to mixed-precision QAT training. Second, high-performance quan-

tization models (such as VGG/ResNet variants) trained on datasets like CIFAR-10/100 and DVS-CIFAR10 will be released. These models are expected to achieve state-of-the-art compression ratios (e.g., average bit width less than 3 bits) while keeping accuracy loss within 1.5% of the full-precision baseline. Furthermore, the project will provide a detailed hardware efficiency analysis report, quantitatively demonstrating the advantages of mixed-precision solutions over traditional uniform quantization solutions in terms of MAC/AC operands and memory bandwidth through energy-accuracy tradeoff curves and resource usage comparison tables. Finally, a high-quality dissertation and a code repository containing all experimental reproduction scripts will be submitted, ensuring the scientific rigor and reproducibility of the research conclusions.

4. Research Progress

4.1 Preliminary results

The project's hardware and software infrastructure has been successfully deployed. The software stack is built on the PyTorch deep learning framework and SpikingJelly (a dedicated spiking neural network computation library). This environment is configured with an automatic differentiation engine to support surrogate gradient computation and integrates efficient CUDA-accelerated neuron models (such as LIF/PLIF), providing efficient parallel computing support for time-step-based SNN simulations.

Currently, we are in the critical stage of establishing a robust benchmark model (corresponding to Phase 0 of the Task Plan) to ensure the fairness and persuasiveness of subsequent comparative experiments of mixed-precision methods. In terms of model architecture, VGG-16 and ResNet-18 have been clearly selected as the main backbone networks. These two architectures are widely used in relevant literature and are easy to compare horizontally. In terms of algorithm benchmark, after in-depth research, the 2-bit and 4-bit uniform quantization schemes of MINT (Yin et al., 2023) [3] have been formally selected as the core control group because they

represent the current state-of-the-art (SOTA) level of uniform quantization SNNs and have no additional pruning interference. The current focus is on integrating the quantization operator of MINT into the code library. At the same time, we are fine-tuning the full-precision (FP32) baseline. The goal is to reproduce the standard performance reported in the literature (e.g., VGG-16 achieves an accuracy of about 92% on CIFAR-10) to establish a reliable performance upper limit.

Data preparation is proceeding as planned. The static datasets CIFAR-10 and CIFAR-100 have been preprocessed, and standard data augmentation pipelines, including Cutout and AutoAugment, have been implemented to prevent overfitting. Addressing the neuromorphic computational characteristics emphasized in the Task Plan, event-driven datasets such as N-MNIST and DVS-CIFAR10 are currently being processed. Integrating loaders that convert event streams to time-step frames are being debugged and validated to ensure that the temporal dimension information of the input data is correctly preserved and utilized during training.

References

[1] Akopyan, F., Sawada, J., Cassidy, A., Alvarez-Icaza, R., Arthur, J., Merolla, P., Imam, N., Nakamura, Y., Datta, P., Nam, G.-J., Taba, B., Beakes, M., Brezzo, B., Kuang, J. B., Manohar, R., Risk, W. P., Jackson, B.Modha & Modha, D. S. TrueNorth: Design and Tool Flow of a 65 mW 1 Million Neuron Programmable Neurosynaptic Chip. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, **2015**, 34(10), 1537.

[2] Davies, M., Srinivasa, N., Lin, T.-H., Chinya, G., Cao, Y., Choday, S. H., Dimou, G., Joshi, P., Imam, N., Jain, S., Liao, Y., Lin, C.-K., Lines, A., Liu, R., Mathaikutty, D., McCoy, S., Paul, A., Tse, J., Venkataramanan, G., Weng, Y.-H., Wild, A., Yang, Y.Wang & Wang, H. Loihi: A Neuromorphic Manycore Processor with On-Chip Learning. *IEEE Micro*, **2018**, 38(1), 82.

[3] Yin, R., Li, Y., Moitra, A.Panda & Panda, P. MINT: Multiplier-less INTeger Quantization for Energy Efficient Spiking Neural Networks. *arXiv preprint arXiv:2305.09850*, **2023**, https://arxiv.org/abs/2305.09850.

[4] Wei, W., Liang, Y., Belatreche, A., Xiao, Y., Cao, H., Ren, Z., Wang, G., Zhang, M., Yang & Yang, Y.Q-SNNs: Quantized Spiking Neural Networks., **2025**, https://arxiv.org/abs/2406.13672.

[5] Wei, W., Zhang, M., Zhou, Z., Belatreche, A., Shan, Y., Liang, Y., Cao, H., Zhang, J., Yang & Yang, Y.QP-SNN: Quantized and Pruned Spiking Neural Networks., **2025**, https://arxiv.org/abs/2502.05905.

[6] Hu, Y., Zheng, Q., Jiang, X.Pan & Pan, G. Fast-SNN: Fast Spiking Neural Network by Converting Quantized ANN. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, **2023**, 45(12), 14546.

[7] Li, C., Ma, L.Furber & Furber, S. Quantization Framework for Fast Spiking Neural Networks. *Frontiers in Neuroscience*, **2022**, 16, , http://dx.doi.org/10.3389/fnins.2022.918793.

[8] Yao, M., Hu, J., Zhou, Z., Yuan, L., Tian, Y., Xu, B., Li & Li, G.Spike-driven Transformer., **2023**, https://arxiv.org/abs/2307.01694.

[9] Dong, Z., Yao, Z., Gholami, A., Mahoney, M.Keutzer & Keutzer, K. HAWQ: Hessian AWare Quantization of Neural Networks With Mixed-Precision. In *2019 IEEE/CVF International Conference on Computer Vision (ICCV)*. **2019**, 293–302.

[10] Moshruba, A., Snyder, S., Poursiami, H.Parsa & Parsa, M. On the Privacy-Preserving Properties of Spiking Neural Networks with Unique Surrogate Gradients and Quantization Levels. In *2025 International Joint Conference on Neural Networks (IJCNN)*. **2025**, 1–8.

[11] Na, B., Mok, J., Park, S., Lee, D., Choe, H., Yoon & Yoon, S.AutoSNN: Towards Energy-Efficient Spiking Neural Networks., **2022**, https://arxiv.org/abs/2201.12738.